

PADERBORN LIPPSTADT, GermanyWMO: **104260**Lat: **51.614N**Lon: **8.616E**Elev: **699**StdP: **14.33**Time Zone: **1.00 (EUC)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	21.1	9.0	8.9	21.6	13.6	11.3	23.6	29.3	44.5	26.5	45.4	7.1	150	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	86.1	66.2	82.2	64.6	78.6	63.4	67.9	80.7	66.5	78.2	64.9	75.1	8.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	92.3	71.5	62.5	86.9	70.1	60.9	82.0	68.9	32.7	80.3	31.5	77.4	30.3	75.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	20.2	18.0	DB	11.2	93.8	5.8	3.0	7.0	95.9	3.7	97.7	0.4	99.4	-3.8	101.5	(4)
(5)			WB	10.2	71.7	5.3	1.8	6.4	73.1	3.3	74.1	0.4	75.2	-3.5	76.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	36.0	36.3	41.7	49.8	56.0	61.5	65.1	64.7	58.7	51.2	43.6	39.0
	DBStd	12.51	6.89	8.10	7.83	7.40	7.10	6.45	6.56	5.99	6.21	6.80	6.58	7.28
	HDD50	1833	435	383	271	95	23	1	0	0	3	69	209	344
	HDD65	5560	899	803	721	459	290	142	81	79	210	428	643	805
	CDD50	1971	0	1	14	87	210	347	468	455	263	106	16	4
	CDD65	225	0	0	0	1	12	38	84	69	20	1	0	0
	CDH74	1918	0	0	0	19	104	348	758	553	134	2	0	0
	CDH80	557	0	0	0	0	15	89	252	178	23	0	0	0
Wind	WSAvg	8.6	10.4	9.8	9.6	8.0	7.4	7.1	7.4	7.3	7.5	8.9	9.2	10.7
Precipitation	PrecAvg	33.3	2.8	2.4	2.5	2.0	2.6	3.0	3.4	3.3	2.8	2.6	3.0	3.0
	PrecMax	45.9	6.4	4.8	5.6	3.7	5.4	4.8	5.7	8.2	5.9	7.6	5.1	6.3
	PrecMin	26.1	0.3	0.7	0.3	0.2	1.0	1.2	1.0	1.0	0.4	0.7	0.0	1.1
	PrecStd	5.4	1.5	1.2	1.4	0.9	1.2	0.9	1.3	1.6	1.5	1.4	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	60.4	66.1	77.0	82.1	87.6	91.6	91.4	82.8	71.8	62.5	55.8
		MCWB	49.8	46.1	51.5	58.0	63.8	67.2	67.0	65.9	64.0	60.6	54.7	50.0
	2%	DB	50.1	53.2	60.7	71.4	76.9	82.2	86.3	85.7	78.5	66.6	58.1	53.7
		MCWB	46.3	44.7	49.2	55.3	61.1	65.8	66.1	65.9	63.0	58.0	52.2	48.8
	5%	DB	48.0	50.0	55.8	66.4	73.1	77.3	82.0	80.5	73.3	63.0	55.3	51.6
		MCWB	45.4	45.5	47.6	53.5	58.9	63.6	64.6	64.4	61.3	56.1	50.5	47.2
	10%	DB	44.8	47.9	52.2	62.6	69.5	73.5	77.3	75.5	69.4	60.7	52.1	48.6
		MCWB	42.4	44.0	45.8	52.3	57.4	61.9	63.5	62.7	59.7	55.3	48.1	45.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	49.2	53.3	59.6	67.1	70.0	69.9	69.5	67.0	61.6	55.5	51.9
		MCDB	52.7	51.9	61.9	70.9	77.9	81.3	84.8	83.6	77.0	69.1	60.1	54.8
	2%	WB	47.1	47.6	50.5	56.9	63.5	67.3	67.7	67.3	64.4	59.1	52.8	49.7
		MCDB	49.3	51.1	57.4	67.8	72.5	78.6	81.0	80.2	74.9	64.8	57.2	52.7
	5%	WB	44.8	45.5	49.0	55.0	60.7	65.1	66.2	65.6	62.4	57.3	50.9	47.8
		MCDB	46.9	49.5	54.9	63.8	69.4	74.7	78.0	76.9	71.5	62.5	54.8	51.2
	10%	WB	42.5	43.2	46.7	52.9	58.2	63.0	64.5	63.8	60.5	55.5	48.7	45.6
		MCDB	44.3	47.2	51.5	61.4	66.9	71.5	74.7	73.2	67.5	60.2	52.1	48.5
Mean Daily Temperature Range	5% DB	MDBR	7.5	10.0	12.3	16.2	16.6	17.1	16.5	15.9	14.2	11.5	8.8	8.1
		MCDBR	10.1	14.2	18.5	22.7	22.5	23.6	23.9	23.4	20.4	15.1	12.3	9.2
		MCWBR	8.7	9.4	10.7	11.6	11.3	11.4	9.4	9.6	9.8	9.3	8.8	7.4
	5% WB	MCDBR	8.6	12.3	15.6	19.2	19.9	20.5	20.7	19.1	18.1	14.0	11.0	8.8
		MCWBR	8.3	9.3	9.7	10.5	11.1	11.1	9.2	9.3	9.5	9.2	8.5	7.8
Clear-Sky Solar Irradiance	taub	0.309	0.332	0.365	0.401	0.405	0.405	0.416	0.411	0.379	0.350	0.330	0.306	
	taud	2.405	2.369	2.302	2.242	2.261	2.280	2.252	2.290	2.368	2.437	2.417	2.397	
	Ebn at Noon	224	250	261	265	269	270	264	259	254	240	213	204	
	Edh at Noon	19	26	33	39	41	40	41	37	31	24	19	16	
All-Sky Solar Radiation	RadAvg	217	431	780	1251	1510	1640	1558	1328	957	558	250	163	
	RadStd	31	69	103	154	162	147	210	122	131	57	37	19	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page